Classroom

-class_name : std::string
-students : std::vector<Student>

+Classroom()
+Classroom(class_name: std::string)
+Classroom(class_name: std::string, students: std::vector<Student>)
+getClassName() : std::string
+getStudents() : std::vector<Student>
+setClassName(class_name: std::string) : void
+addStudent(student: Student) : void
+removeStudent(fac_num: std::string) : void
+searchByFacultyNumber(fac_num: std::string) : int
+sortStudents() : void
+filterStudentsByGPA(gpa_min: float) : std::vector<Student>
+filterStudentsByGPA(gpa_min: float, gpa_max: float) : std::vector<Student>
+printStudentAgeDistribution() : void
+printStudentAgeDistribution(date: Date) : void
+averageClassGPA() : float
+isClassNameValid(class_name: std::string) : bool
+printStudentInfo(student: Student) : void
+printClassStudents() : void
+isFileExtensionValid(file_path: std::string) : bool
+saveData(file_path: std::string) : void
+loadData(file_path: std::string) : void

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Student

-name : std::string
-faculty_number : std::string
-average_gpa : float
-birth_date : Date

+Student()
+Student(name: std::string, fac_num: std::string, avg_gpa: float, birthday: Date)
+getName() : std::string
+getFacultyNumber() : std::string
+getAverageGPA() : float
+getBirthDate() : Date
+setName(name: std::string) : void
+setFacultyNumber(fac_num: std::string) : void
+setAvgGPA(avg_gpa: float) : void
+setBirthDate(birthday: Date) : void
+getAge() : int
+getAge(date: Date) : int
+isFacultyNumberValid(fac_num: std::string) : bool
+operator<(other_student: Student) : bool
+printStudent() : void

Date

-day : int
-month : int
-year : int
-DAYS_IN_EACH_MONTH_NON_LEAP : int[12]

+Date()
+Date(d: int, m: int, y: int)
+getDay() : int
+getMonth() : int
+getYear() : int
+printDate() : void
+setDate(d: int, m: int, y: int) : void
+isLeapYear(y: int) : bool
+getDaysInMonth(m: int, y: int) : int
+isDateValid(d: int, m: int, y: int) : bool
+getCurrentYear() : int
+getCurrentMonth() : int
+getCurrentDay() : int